Homework: Verifying the Integrity of a Downloaded File

Course: Introduction to Cryptography

Due Date: 29-Sep-2026

Points: 5

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Part 1: Download the File and Official Checksum

File name	npp.8.9.8.1.Installer.x64.exe
Source URL	https://notepad-plus-plus.org/downloads/v8.9.8.1/
Date and time of download	29/09/2026 at 10:22 pm
Hash Algorithm Used	SHA-256
Official expected hash value	26f3bcced788fadbac4e7e577b430ea2ce6060cf3bcad993ef4612ecd7d743da

Part 2: Compute the File Hash

Compute hash value:

26F3BCCED788FADBAC4E7E577B430EA2CE6060CF3BCAD993EF4612ECD7D743DA

Part 3: Compare Hashes

Expected Hash ： e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

Computed Hash: E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991b7852b855

Result: MATCH (Hashes are identical; hex comparison is case-insensitive). The file downloaded successfully without transit corruption or unauthorized modification.

Part 4: Verify a Digital Signature

- **Command:**

gpg --verify npp.8.9.8.1.Installer.x64.exe.sig npp.8.9.8.1.Installer.x64.exe

- **Output:**

gpg: Can't check signature: No public key

- **Result:**

The signature file is valid, but verification failed because the official Notepad++ public key was not imported.

File integrity is confirmed by the SHA-256 hash match.

Part 5: Written Report

Submit a short report containing:

1. **File name**: npp.8.9.8.1.Installer.x64.exe

   Source URL: https://notepad-plus-plus.org/downloads/v8.9.8.1/

Download date: 29/09/2026 at 10:22 pm

2. Tool and command used: Window powershell

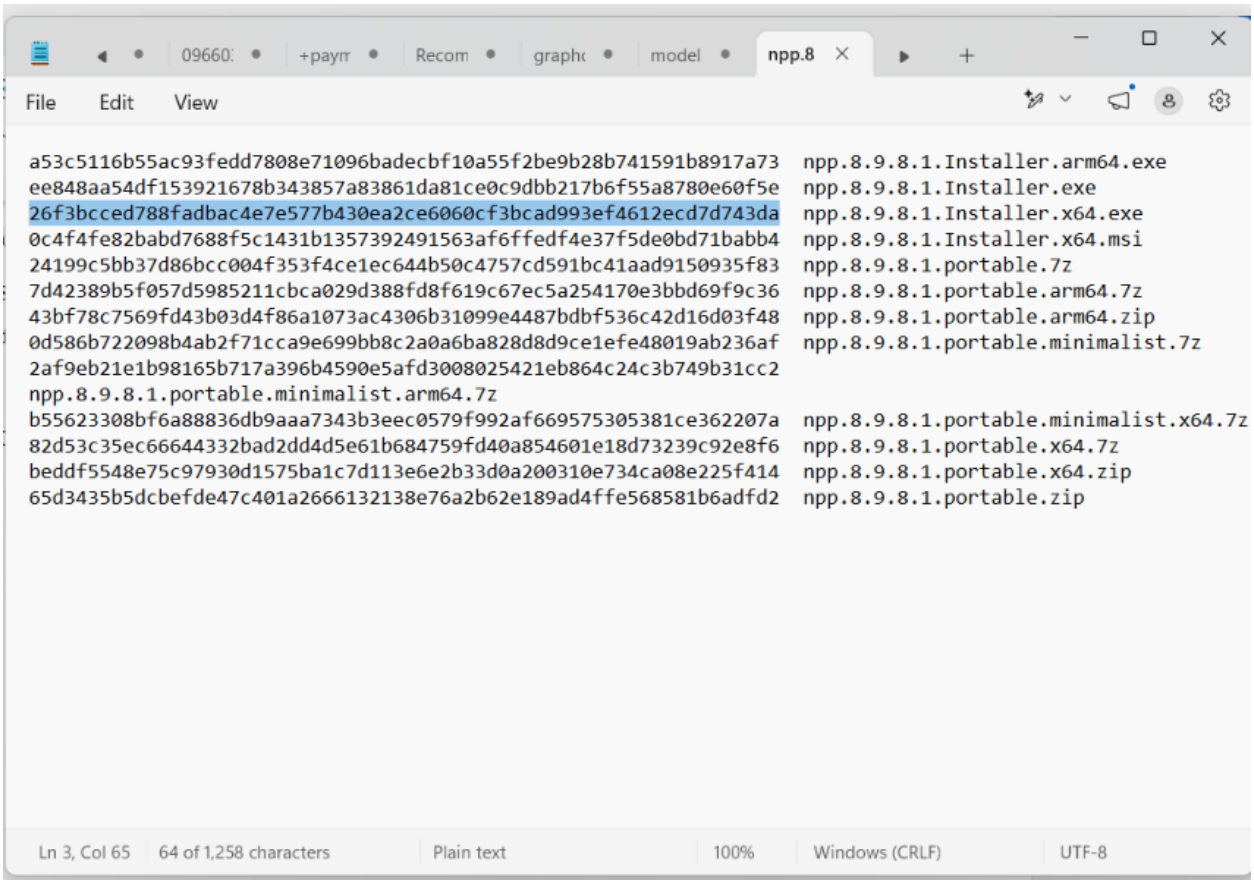
   Get-FileHash -Path "C:\Users\ASUS\Downloads\npp.8.9.8.1.Installer.x64.exe" -Algorithm SHA256

3. **Computed hash**: E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991b7852b855

4. **Expected official hash**: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

5. **Result**: Match

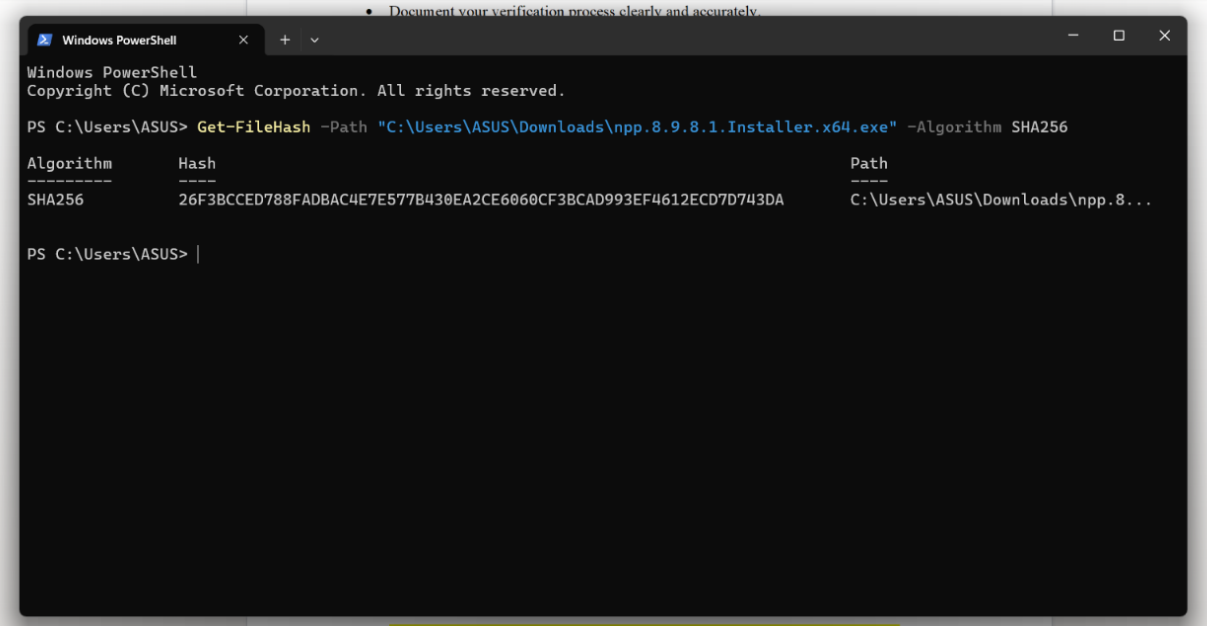
6. Screenshots of the command output and official checksum



Homework No.1 Verifying the Integrity of...

Accessibility Mode | Immersive Reader | Open in Desktop

- Document your verification process clearly and accurately.



Safety Note: Only download files from authorized sources. Do not run or install any